

Module 2

Data Structures and Algorithms

Exercise 1: Inventory Management System

Code:

```
Main.java
1 import java.util.*;
2 class Product {
3     int id;
4     String name;
5     int qty;
6     double price;
7
8     Product(int id, String name, int qty, double price) {
9         this.id = id;
10        this.name = name;
11        this.qty = qty;
12        this.price = price;
13    }
14 }
15 public class Main {
16     public static void main(String[] args) {
17         HashMap<Integer, Product> items = new HashMap<>();
18
19         items.put(101, new Product(101, "Laptop", 10, 50000));
20         items.put(102, new Product(102, "Mouse", 20, 500));
21
22         items.put(102, new Product(102, "Mouse", 30, 550));
23
24         items.remove(101);
25
26         for (Product p : items.values()) {
27             System.out.println(p.id + " " + p.name + " " + p.qty + " " + p.price);
28         }
29     }
30 }
```

Output:

```
102 Mouse 30 550.0

...Program finished with exit code 0
Press ENTER to exit console.
```

Exercise 2: E-Commerce Search

Code:

```

Main.java :
1 class Product {
2     int id;
3     String name;
4     String category;
5
6     Product(int id, String name, String category) {
7         this.id = id;
8         this.name = name;
9         this.category = category;
10    }
11 }
12 public class Main {
13
14     static int linear(Product arr[], String key) {
15         for (int i = 0; i < arr.length; i++) {
16             if (arr[i].name.equalsIgnoreCase(key))
17                 return i;
18         }
19         return -1;
20     }
21
22     static int binary(Product arr[], String key) {
23         int l = 0;
24         int h = arr.length - 1;
25
26         while (l <= h) {
27             int m = (l + h) / 2;
28
29             int x = arr[m].name.compareToIgnoreCase(key);
30
31             if (x == 0)
32                 return m;
33             else if (x < 0)
34                 l = m + 1;
35             else
36                 h = m - 1;
37         }
38
39         return -1;
40     }
41
42     public static void main(String[] args) {
43
44         Product p[] = {
45             new Product(1, "Camera", "Electronics"),
46             new Product(2, "Laptop", "Electronics"),
47             new Product(3, "Mobile", "Electronics"),
48             new Product(4, "Watch", "Accessories")
49         };
50
51         System.out.println(linear(p, "Mobile"));
52         System.out.println(binary(p, "Mobile"));
53     }
54 }

```

Output:

```

2
2

...Program finished with exit code 0
Press ENTER to exit console.

```

Exercise 3: Sorting Customer Orders

Code:

```
Main.java
1 class Order {
2     int id;
3     String customer;
4     double amount;
5     Order(int id, String customer, double amount) {
6         this.id = id;
7         this.customer = customer;
8         this.amount = amount;
9     }
10 }
11 public class Main {
12     static void bubble(Order a[]) {
13         for (int i = 0; i < a.length - 1; i++) {
14             for (int j = 0; j < a.length - i - 1; j++) {
15                 if (a[j].amount > a[j + 1].amount) {
16                     Order t = a[j];
17                     a[j] = a[j + 1];
18                     a[j + 1] = t;
19                 }
20             }
21         }
22     }
23     public static void main(String[] args) {
24         Order a[] = {
25             new Order(1, "Ravi", 4000),
26             new Order(2, "Anu", 1500),
27             new Order(3, "Kiran", 6000) };
28         bubble(a);
29         for (Order x : a)
30             System.out.println(x.id + " " + x.customer + " " + x.amount);
31     }
}
```

Output:

```
2 Anu 1500.0
1 Ravi 4000.0
3 Kiran 6000.0

...Program finished with exit code 0
Press ENTER to exit console.
s
```

Exercise 4: Employee Management System

Code:

```
Main.java :
1 - class Employee {
2     int id;
3     String name;
4     String post;
5     double salary;
6 - Employee(int id, String name, String post, double salary) {
7     this.id = id;
8     this.name = name;
9     this.post = post;
10    this.salary = salary;
11 }
12 }
13 - public class Main {
14 -     public static void main(String[] args) {
15         Employee e[] = new Employee[5];
16         e[0] = new Employee(1, "Ravi", "Manager", 50000);
17         e[1] = new Employee(2, "Sita", "Developer", 40000);
18 -         for (int i = 0; i < e.length; i++) {
19             if (e[i] != null)
20                 System.out.println(e[i].id + " " + e[i].name);
21         }
22         int searchId = 2;
23 -         for (int i = 0; i < e.length; i++) {
24             if (e[i] != null && e[i].id == searchId)
25                 System.out.println("Found " + e[i].name);
26         }
27
28         e[1] = null;
29     }
30 }
```

Output:

```
1 Ravi
2 Sita
Found Sita

...Program finished with exit code 0
Press ENTER to exit console.
```

Exercise 5: Task Management System
Code:

```

1 class Task {
2     int id;
3     String name;
4     String status;
5     Task next;
6     Task(int id, String name, String status) {
7         this.id = id;
8         this.name = name;
9         this.status = status;
10    }
11 }
12 public class Main {
13     static Task head;
14     static void add(int id, String name, String status) {
15         Task t = new Task(id, name, status);
16         if (head == null) {
17             head = t;
18             return;
19         }
20         Task temp = head;
21
22         while (temp.next != null)
23             temp = temp.next;
24
25         temp.next = t;
26     }
27
28     static void display() {
29         Task temp = head;
30
31         while (temp != null) {
32             System.out.println(temp.id + " " + temp.name + " " + temp.status);
33             temp = temp.next;
34         }
35     }
36
37     public static void main(String[] args) {
38
39         add(1, "Coding", "Pending");
40         add(2, "Testing", "Completed");
41
42         display();
43     }
44 }

```

Output:

```

1 Coding Pending
2 Testing Completed

...Program finished with exit code 0
Press ENTER to exit console.

```

Exercise 6: Library Management System

Code:

```

Main.java
1 class Book {
2     int id;
3     String title;
4     String author;
5
6     Book(int id, String title, String author) {
7         this.id = id;
8         this.title = title;
9         this.author = author;
10    }
11 }
12 public class Main {
13
14     static int linear(Book b[], String key) {
15         for (int i = 0; i < b.length; i++) {
16             if (b[i].title.equalsIgnoreCase(key))
17                 return i;
18         }
19         return -1;
20     }
21
22     static int binary(Book b[], String key) {
23         int l = 0;
24         int h = b.length - 1;
25
26         while (l <= h) {
27             int m = (l + h) / 2;
28
29             int x = b[m].title.compareToIgnoreCase(key);
30
31             if (x == 0)
32                 return m;
33             else if (x < 0)
34                 l = m + 1;
35             else
36                 h = m - 1;
37         }
38
39         return -1;
40     }
41
42     public static void main(String[] args) {
43
44         Book b[] = {
45             new Book(1, "C", "Dennis"),
46             new Book(2, "Java", "James"),
47             new Book(3, "Python", "Guido")
48         };
49
50         System.out.println(linear(b, "Java"));
51         System.out.println(binary(b, "Java"));
52     }
53 }

```

Output:

```

1
1

...Program finished with exit code 0
Press ENTER to exit console.

```

Exercise 7: Financial Forecasting

Code:

```
Main.java
1 public class Main {
2
3     static double predict(double amount, double rate, int years) {
4
5         if (years == 0)
6             return amount;
7
8         return predict(amount * (1 + rate), rate, years - 1);
9     }
10
11     public static void main(String[] args) {
12
13         double amount = 10000;
14         double rate = 0.10;
15         int years = 5;
16
17         System.out.printf("%.2f", predict(amount, rate, years));
18     }
19 }
```

Output:

```
16105.10

...Program finished with exit code 0
Press ENTER to exit console.
```